



EPICODE

Frontend Programming

Project Overview



You are required to develop a React application that demonstrates your understanding of modern React concepts, including state management, routing, asynchronous operations, and API integration. The project should be fully functional and well-structured, showcasing good coding practices

This practical project accounts for 50% of the final grade; the remaining 50% will be assessed through an oral examination



General Requirements

Project Theme

You are free to choose the theme of your application (e.g., an e-commerce store, a task manager, a social media dashboard, a movie database, etc.)



General Requirements

Components

Break the UI into reusable and well-structured components and use conditional rendering and component composition where possible



General Requirements

State Management

Use `useState` and `useEffect` hooks to manage component state and side effects. Use `Redux` for global state management and `Thunk` for handling asynchronous actions



General Requirements

Pages and Routing

The application should include at least six distinct pages, covering essential areas such as a homepage, a dashboard for administrators or content creators, one or more listing pages (with filters and paginated content), and at least one detail page that shows information about a specific item(e.g., /product/42 or /post/123). Use React Router for navigation, including dynamic routing for these detail pages



General Requirements

Users

It must simulate the presence of users by implementing a login system (fake authentication is acceptable). Additionally, it should support at least two user roles (e.g., regular user and admin) with different views. For example, admins might have access to management features or additional content sections



General Requirements

API Usage

The application must consume external API for dynamic content (e.g., fetching data from a public API like OpenWeather, TMDB, JSONPlaceholder, etc.) or custom API with Json Server (or both)



General Requirements

Forms

Include at least four controlled forms with validation. These forms should mimic real-world functionality and be closely tied to the application's features. For example:

- In a marketplace app, an admin user should be able to create, update, and delete products through a form
- In a social media app, a user should be able to create, edit, and delete their posts using forms that handle user input

Make sure that each form is fully functional, and incorporates validation for required fields, proper formats, and error handling



General Requirements

Submission Requirements

The project must be hosted on GitHub or submitted via .zip file on LMS, including everything needed to run the application and a clear README.md explaining: project overview, running instructions, features, technologies used



Optional Features

Use CSS, Bootstrap, Tailwind, or another styling library

Feel free to use additional libraries to improve UX/UI

Optimize performance where applicable

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✗ All general requirements are mandatory. Failure to meet these requirements will result in penalties in the final evaluation

✓ The optional features (bonus points) are not required but can contribute to a higher score if implemented effectively

This assignment is designed to give you creative freedom while ensuring that you demonstrate proficiency in key React concepts. Choose a project idea that interests you and make it your own while following the guidelines

Good luck and happy coding! 🚀



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